Copper and Cast Iron

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Q I live in a 1920s Wardman row house in Washington, D.C. The house has a cast-iron radiator heating system. I hired a local contractor to do some bathroom renovations. It was necessary to remove a large cast-iron radiator from the bathroom to make room for the remodel. The contractor hired a plumber who installed a copper, baseboard-style radiator. It looked great! But the problem is that it only ekes out a minimal amount of heat. All of the other cast-iron radiators in the house sizzle with heat, and this copper baseboard heater is tepid at best. I called the plumber to come out to address the problem. He re-bled the system and said that it was fixed and working correctly. However, we noticed no appreciable change whatsoever.

What can be done to resolve our problem? It's December, and we need heat in the bathroom!

Tempted to Stay in Bed
Washington, D.C.

A. This is a common problem with renovations. Here is why.

Many hot water heating systems are controlled by a single thermostat that turns the heat completely on or off buildingwide. When the thermostat is "calling" for heat, the pump moves hot water, typically 80°C–85°C (176°F–185°F) all heating season, through the radiators and baseboard. When it is off, the pump stops and the water stops moving.

When the system is on, the hundreds of kilos of cast iron in those radiators heat up, as does the half kilo or so of copper in the baseboard. When the thermostat is satisfied, the pump turns off, leaving many liters of hot water in the cast-iron radiators, and only a few cubic centimeters in the baseboard. The radiators continue to put out heat for 20 or 30 minutes, while the baseboard cools down in a few minutes. This results in terribly uneven heat.

This resulted in a rule of thumb that cast-iron and copper shouldn't be used together, which has been simplified into a belief that they can't work together. It is easy to get them to work together, but not the way the heating system is installed in your home.

One of the advantages hot-water heat has over scorched-air heat is that it is easy to make each room a control zone. This avoids the overheating associated with one uncomfortable person in one cold room, replaces the fan energy with a small pump load, and eliminates the air leakage through the walls and every infiltration/exfiltration path that is driven by uneven air pressures caused by fan operation. While this is widely considered impractical or impossible in the United States, it is hard to find a room in Europe that does not have its own hot-water heating thermostat. They work there, with the same laws of physics that we have here.

The first thing to do is to check to see if your house is already piped with a separate supply and return to each radiator. Turn off one radiator and see if any other radiators turn off, which they shouldn't. (If another radiator turns off, you don't have individual controls on each radiator, and the only way to get individual control would be through repiping. Fortunately, most older houses with cast iron radiators were piped with a separate supply and return to each radiator.) Then get a plumber to remove each radiator's shutoff valve and install a thermostatic radiator valve in its place. It is important that the plumber follow the instructions for installing these valves—flow should follow the arrow on the valve, and the control should be on the side of the valve, not on the top where it will "see" hot air from the pipe.

Then get a plumber or heating contractor to install an outdoor reset control on your boiler. The reset control will turn the pump on when the outdoor temperature drops below 12°C or 13°C (54°F or 55°F) or so, and leave it on constantly until the weather warms up (yes, the pump uses energy, but only about 60 to 100 watts). It will also change the temperature of the water going to the radiators—warm in cool weather, hot in cold weather. (The best outdoor reset controls have a night setback option that lowers the water temperature at night, cooling down the whole house regardless of what the individual thermostats are set for.) This is like going from controlling the speed of your car by turning the key on and off to using the gas pedal. You will have even heat all season long, at whatever temperature you set the thermostats for.

Since the system stays on continuously, the problem with mixing cast-iron and copper heaters will be eliminated. If the copper heater is not large enough, it will still have to be replaced with something larger. Calculations could show if it is large enough—as a joke, you could ask your contractor to fax over his calculations. But regardless of what size heater you need, the benefits of an outdoor reset control and individual thermostats are compelling, so you should install them in any case.